GI Pathology 1.1 Structural Esophageal Disorders & GERD



1. **False mucosal teeth** : a false diverticulum is an outpouching of mucosa and submucosa only (versus a true diverticulum which is outpouching of mucosa, submucosa, muscularis propria, and adventitia)
2. **Zany Zenker's hoodie** : Zenker diverticulum is a false diverticulum in the posterior upper esophagus
3. **"kIds go for enjoyment"** : Zenker's diverticulum is due to weakness of the cricopharyngeus muscle in the Killian triangle (an area of the hypopharynx)
4. **Eating burning pizza** : Zenker's diverticulum may present with odynophagia (painful swallowing)
5. **Stinky cheese** : Zenker's diverticulum may present with halitosis (bad breath)
6. **Regurgitating food** : Zenker's diverticulum may present with regurgitation
7. **Protruding "HIT-A-Mouse" game**: in a HIATAL HERNIA, the stomach herniates into the thorax through the esophageal hiatus
8. **Sliding mouse-spring junction** : in a SLIDING hernia, the gastroesophageal junction herniates through the esophageal hiatus (a type of HIATAL HERNIA)
9. **2 protruding mice** : in a PARAESOPHAGEAL hernia, the gastric cardia herniates through the esophageal hiatus (sneaking up beside a normally placed esophagus)(a type of HIATAL HERNIA)
10. **Refluxing acid** : HIATAL HERNIAs present with gastroesophageal reflux (GERD)
11. **Marshmallow pit** : Mallory-Weiss tears (mucosal tears in the proximal stomach and distal esophagus)
12. **Linear scratches on distal slide** : Mallory-Weiss tears consist of linear longitudinal lacerations in the mucosa of the distal esophagus and proximal stomach
13. **Drinking at marshmallow pit** : Mallory-Weiss tears are often due to alcohol abuse (increased intragastric pressure from vomiting)
14. **"Wrenching" at marshmallow pit**: Mallory-Weiss tears (and Boerhaave syndrome) are caused by increased intragastric pressure during retching and vomiting
15. **Coughing-up red drink** : Mallory-Weiss tears present with mild to moderate hematemesis
16. **Spilling red drink** : Mallory-Weiss tears can present with moderate hematemesis
17. **Kick to the stomach** : Mallory-Weiss tears can cause epigastric pain
18. **Ruptured bounce house** : in Boerhaave syndrome, there is transmural rupture of the esophagus
19. **"Wrenching" at marshmallow pit**: Mallory-Weiss tears (and Boerhaave syndrome) are caused by increased intragastric pressure during retching and vomiting
20. **PeriSCOPE** : Boerhaave syndrome can be caused by esophageal endoscopy
21. **Weakened bounce house** : conditions that weaken the esophageal wall (e.g. infection, inflammation, structural defects) are risk factors for Boerhaave syndrome
22. **Air-puff induced chest injury** : Boerhaave syndrome causes severe sharp chest pain
23. **Spilling red juice** : Boerhaave syndrome causes severe hematemesis
24. **CRUNCHY chip** : Boerhaave syndrome can cause subcutaneous emphysema (palpation of chest causes "crunchy" sound
25. **Wet pleural shirt** : Boerhaave syndrome can cause pleural effusions (due to gastric contents in pleural space)
26. **SEPTIC manhole cover and flame bandana** : Boerhaave syndrome can cause fever, sepsis, and septic shock

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27. **Gargling** : gastroesophageal reflux disease (GERD) (an imbalance between LES tone and intragastric pressure causes reflux of gastric acid into the esophagus)
28. **Loose drawstring** : decreased LES tone can cause GERD
29. **Smoker** : smoking can cause GERD (due to decreased LES tone)
30. **Alcoholic** : alcohol use can cause GERD (due to decreased LES tone)
31. **Coffee** : caffeine can cause GERD (due to decreased LES tone)
32. **Pregnant woman** : pregnancy can cause GERD (due to decreased LES tone and increased intragastric pressure)
33. **Obese guy** : obesity can cause GERD (due to increased intragastric pressure)
34. **Punching chest** : GERD can present with noncardiac chest pain such as heartburn or indigestion
35. **Coughing in bed** : GERD can present with nocturnal cough
36. **Inhaler** : GERD can present with nocturnal asthma
37. **Choking throat** : GERD can present with dysphagia (difficulty swallowing
38. **Burning hole** : GERD can progress to erosive esophagitis or esophageal ulcers
39. **Clutching throat** : erosive esophagitis due to GERD can present with odynophagia (painful swallowing)
40. **Mucosal cake with pink and blue sprinkles** : severe GERD causes esophagitis (on histology, appears as eosinophils and lymphocytes infiltrating the mucosa)
41. **Tall lower cake layer with thick frosting** : severe GERD causes elongation of the papillae of the lamina propria and hypertrophy of the basal cells of the mucosa
42. **Barrett's bear** : GERD can lead to Barrett's esophagus (intestinal metaplasia of the distal esophagus)
43. **Squamous tiles** : the esophagus is normally lined with stratified squamous epithelium
44. **Columnar stage** : intestinal metaplasia of Barrett's esophagus consists of columnar epithelium
45. **Scattered mucus cups** : intestinal metaplasia of Barrett's esophagus contains goblet cell hyperplasia
46. **Metal intestines** : intestinal metaplasia (Barrett's esophagus)
47. **Ascending red velvety curtains**: in Barrett's esophagus, patches of red velvety mucosa (columnar epithelium) extend upward from the gastroesophageal junction
48. **Dirty dysplastic floor** : dysplasia is a precancerous condition (compared to metaplasia, which is benign transformation of one cell type into another)
49. **Cancer crab** : Barrett's esophagus (METAplasia) can progress to DYSplasia, and eventually cancer (esophageal adenocarcinoma)
50. **Parasol** : GERD is treated with proton pump inhibitors (PPIs, suffix "prasol") or H2 blockers (both reduce acid production in the stomach)
51. **Eosinophilic bilobed sling-shot** : eosinophilic esophagitis presents similarly to severe GERD (dysphagia, chest/upper abdominal pain after eating
52. **Slingshot kid** : eosinophilic esophagitis often occurs in childhood (unlike GERD, which usually occurs in adults
53. **Pink eosinophilic granules** : eosinophilic esophagitis involves epithelial infiltration of eosinophils throughout the esophagus
54. **Itchy nose and hives** : eosinophilic esophagitis may present along with other atopic symptoms (eczema, allergic rhinitis, asthma, peripheral eosinophilia)
55. **Breaking through parasol** : Eosinophilic esophagitis may be resistant to PPIs